

Chapter 04 - OpenROAD first run

Course authors (Git file)



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- 4 The designs to run



Section 1

Doing this chapter



Doing this chapter

- This chapter is mostly a training.
- We will try to get our first results (GDS files).
- There are example designs available.
- You can start your own design.



Section 2

Download the OpenRoad Flow Scripts from github



Download the OpenRoad Flow Scripts from github

I do a shallow clone of the git repository without the full history.

```
1 cd  
2 git clone --depth 5 --branch vhdl --single-branch https://github.com/fredowski/OpenROAD-flow-scripts.git
```

This creates the local copy in the folder “OpenROAD-flow-scripts”. The difference to the official repository from OpenROAD is the VHDL support that I added.



Section 3

The Makefile



The Makefile

Let's have a look into the Makefile first.



The flow steps

The Makfile (in the /flow directory) contains all the flow steps in the same order we already have seen:

```

1 # =====
2 #
3 # SYNTHESIS
4 #
5 # =====
6 #
7 # PHONY: sv0
8 #
9 # FLOORPLAN
10 #
11 # =====
12 #
13 # PHONY: 34 #
14 #
15 # 35 #
16 # 36 #
17 # 37 #
18 # 38 #
19 # 39 #
20 # 40 #
21 # 41 #
22 # 42 #
23 #
24 # ROUTING
25 #
26 # =====
27 #
28 # 29 #
29 # 30 #
30 # 31 #
31 # 32 #
32 # 33 #
33 # 34 #
34 # 35 #
35 # 36 #
36 # 37 #
37 # 38 #
38 # 39 #
39 # 40 #
40 # 41 #
41 # 42 #
42 # PHONY: route

```

Figure 1: The flow steps in the Makefile



DESIGN_CONFIG

The Makefile starts with the selection of the design to run.

Of interest for this course are the lines regarding to the IHP PDK:

```
1 #DESIGN_CONFIG=./designs/ihp-sg13g2/aes/config.mk
2 DESIGN_CONFIG=./designs/ihp-sg13g2/counter/config.mk
3 #DESIGN_CONFIG=./designs/ihp-sg13g2/ibex/config.mk
4 #DESIGN_CONFIG=./designs/ihp-sg13g2/gcd/config.mk
5 #DESIGN_CONFIG=./designs/ihp-sg13g2/murax/config.mk
6 #DESIGN_CONFIG=./designs/ihp-sg13g2/spi/config.mk
7 #DESIGN_CONFIG=./designs/ihp-sg13g2/riscv32i/config.mk
8 #DESIGN_CONFIG=./designs/ihp-sg13g2/i2c-gpio-expander/config.mk
```

The `counter` example is selected for the next run.



Section 4

The designs to run



counter - 16 bit modulo 65536 counter with enable (VHDL)

- The counter design is included with the OpenROAD-flow-script examples.
- It consists of only a single VHDL file, easy to read.
- The counter is a synchronous 16 bit modulo 65536 counter with enable



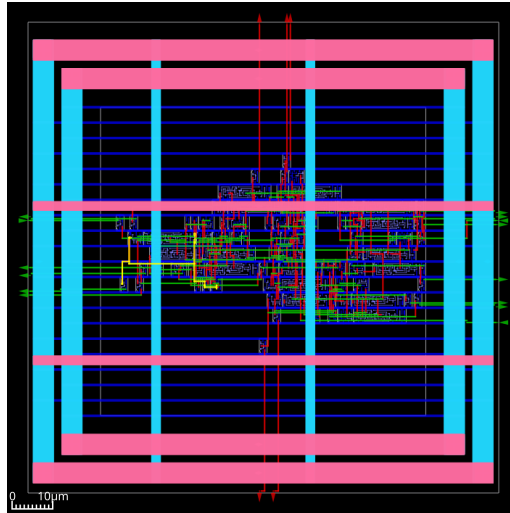


Figure 2: GDS counter



counter files:

```
1 cd flow/designs/src/counter
2 ls -la
3 counter.vhd
```

```
1 cd flow/designs/ihp-sg13g2/counter
2 ls
3 config.mk
4 constraint.sdc
```



Run the flow for counter

Adapt the Makefile by uncommenting the counter design. Run the full flow for the counter example

```
1 cd flow
2 make
```

Open the openroad gui to see the final layout

```
1 cd flow
2 make gui_final
```



Check the results and reports

After each step the design database and reports are saved. The first step is the synthesis step from yosys that creates a verilog netlist with the standard cells. Check the netlist with:

```
1 cd flow/results/ihp-sg13g2/counter/base
2 open 1_2_yosys.v
```

The synthesis report for the example shows the required standard cells and the corresponding area.

```
1 cd flow/reports/ihp-sg13g2/counter/base
2 less synth_stat.txt
```



gcd - greatest common divisor

- The gcd design is included with the OpenROAD-flow-script examples.
- It consists of only a single verilog file, easy to read.
- The gcd design computes the greatest common divisor of two numbers



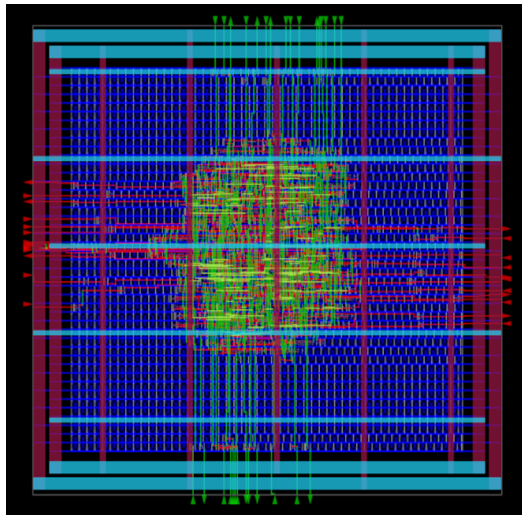


Figure 3: GDS gcd



gcd files:

```
1 cd flow/designs/src/gcd
2 ls
3 gcd.v
4 README.md
5 BUILD.bazel
```

```
1 cd flow/designs/ihp-sg13g2/gcd
2 ls
3 autotuner.json
4 config.mk           (important)
5 constraint.sdc      (important)
6 rules-base.json
```



Run the flow for gcd

Adapt the Makefile by uncommenting the gcd design. Then run the full flow for the gcd example

```
1 cd flow
2 make
```

Open the openroad gui to see the final layout

```
1 cd flow
2 make gui_final
```



Check the gcd results and reports

Check the synthesis netlist with:

```
1 cd flow/results/ihp-sg13g2/gcd/base
2 open 1_2_yosys.v
```

The synthesis report for example show the required standard cells and the corresponding area.

```
1 cd flow/reports/ihp-sg13g2/gcd/base
2 less synth_stat.txt
```



ibex: RISC-V core

- The ibex design is included with the OpenROAD-flow-script examples.
- It consists of many Verilog files, not that easy to read.
- A single run might take more then 30 minutes



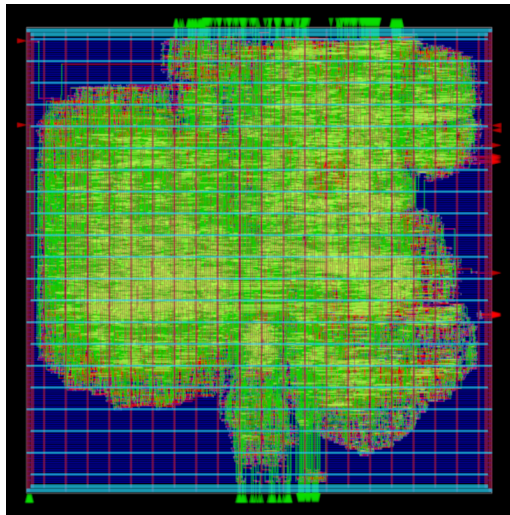


Figure 4: GDS ibex



ibex files:

```

1  src/ibex$ ls
2  ibex_alu.v          ibex_ex_block.v      ibex_register_file_ff.v    prim_ram_1p.v
3  ibex_branch_predict.v  ibex_fetch_fifo.v    ibex_register_file_fpga.v  prim_secded_28_22_dec.v
4  ibex_compressed_decoder.v  ibex_icache.v        ibex_register_file_latch.v  prim_secded_28_22_enc.v
5  ibex_controller.v      ibex_id_stage.v      ibex_wb_stage.v            prim_secded_39_32_dec.v
6  ibex_core.v            ibex_if_stage.v      LICENSE                     prim_secded_39_32_enc.v
7  ibex_counter.v         ibex_load_store_unit.v  prim_badbit_ram_1p.v        prim_secded_72_64_dec.v
8  ibex_cs_registers.v     ibex_multdiv_fast.v    prim_clock_gating.v         prim_secded_72_64_enc.v
9  ibex_csr.v             ibex_multdiv_slow.v    prim_generic_clock_gating.v  prim_xilinx_clock_gating.v
10 ibex_decoder.v          ibex_pmp.v            prim_generic_ram_1p.v        README.md
11 ibex_dummy_instr.v      ibex_prefetch_buffer.v  prim_lfsr.v

```




```
1 | ihp-sg13g2/ibex$ ls
2 | autotuner.json
3 | config.mk
4 | constraint_doe.sdc
5 | constraint.sdc
6 | metadata-base-ok.json
7 | rules-base.json
```



lfsr

- The lfsr design example must be created from scratch.
- The VHDL code is available in the lecture slides and should become a single file.
- The structure of other examples must be copied for this.
- The configuration files must be copied and adapted for this.
- A single run should be very short.



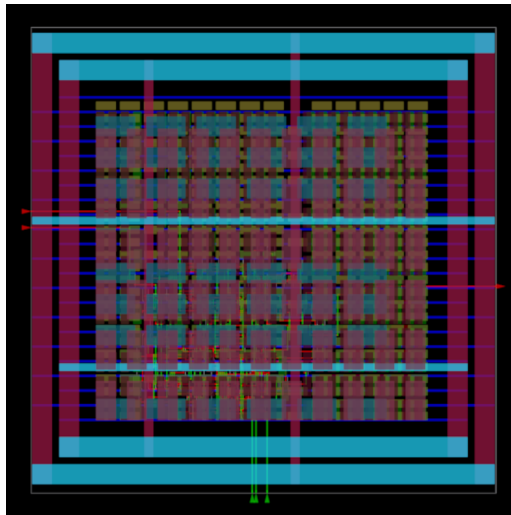


Figure 5: GDS lfsr

